



NIELSEN BOOKSCAN SERVICE– USING TIME SERIES ANALYSIS FOR SALES AND DEMAND FORECASTING

By Shishir. M. Paltanwale

Date: 17th August 2025

TABLE OF CONTENTS

<i>Nielsen BookScan service– Using time series analysis for sales and demand forecasting.....</i>	<i>1</i>
<i>Introduction.....</i>	<i>3</i>
<i>Problem statement.....</i>	<i>3</i>
<i>Overview approach.....</i>	<i>3</i>
<i>Description of the Analysis</i>	<i>4</i>
<i>Explanation of Visualisations</i>	<i>4</i>
<i>Data Visualisation.....</i>	<i>5</i>
<i>Insights</i>	<i>19</i>
<i>Conclusion.....</i>	<i>20</i>
<i>Inferences</i>	<i>20</i>

INTRODUCTION

The Nielsen BookScan service is the world's largest continuous book sales tracking service in the world. The goal of this project was to do a comprehensive analysis of book sales forecasting using Nielsen BookScan data, covering approximately 90% of UK retail print book purchases. The analysis employed classical statistical methods, machine learning techniques, deep learning models, and hybrid approaches on weekly sales data for "The Alchemist" and "The Very Hungry Caterpillar" over a 32-week forecast horizon, leveraging Nielsen BookScan's point-of-sale transactional data for highly accurate forecasting insights.

PROBLEM STATEMENT

Nielsen aims to support small and medium-sized independent publishers by developing a forecasting service that leverages historical sales data to predict future book performance. Independent publishers often lack the infrastructure to make data-driven decisions, leading to risks in stock control and upfront investment.

The challenge is to provide accurate forecasts of sales profiles and the economic lifespan of titles, accounting for seasonal trends, long-term demand, and external influences. By applying advanced models such as: -

1. Auto ARIMA
2. SARIMA
3. XGBoost
4. LSTM and
5. Hybrid approaches

Nielsen seeks to help publishers optimize print volumes, identify titles with longevity, and reduce financial risk.

OVERVIEW APPROACH

The goal was to forecast the **final 32 weeks** of sales for two titles—*The Very Hungry Caterpillar* (VHC) and *The Alchemist* (ALC)—and to compare these forecasts with **monthly** forecasts for the same period ($\approx$ **8 months**). We treated weekly and monthly frequency as complementary lenses: weekly models capture short-term dynamics and promotion effects; monthly models' smooth noise and emphasize seasonal structure.

We implemented three modelling families:

1. **Weekly models**
 - **XGBoost with deseasonalization**: a machine-learning forecaster wrapped in a pipeline that first removes seasonal components (sp=52 weekly) and then predicts on the deseasonalized series with a moving window (grid search over window length = {8,16,32}).
 - **Hybrid parallel model (LSTM + SARIMA)**: an ensemble that linearly combines an LSTM residual forecaster and a SARIMA baseline, with a convex weight α (we tested a setting like $\alpha=(0.5, 0.7, 0.3)$ assigning different weights to SARIMA and LSTM).
2. **Monthly models**
 - **SARIMA via Auto ARIMA**: classical, interpretable time-series model with m=12 to capture annual seasonality, trained on all observations **up to** the start of the forecasting horizon (last 8 months held out).

Accuracy was assessed using **MAE** and **MAPE** on the hold-out horizon. Residual diagnostics (residual plots, ACF, and distribution) were used to sanity-check assumptions (whiteness, remaining seasonality, outliers).

DESCRIPTION OF THE ANALYSIS

The analysis compared classical, machine learning, deep learning, and hybrid models to forecast book sales, with weekly (32 weeks) and monthly (8 months) horizons.

Classical Approach:

- **Auto ARIMA (SARIMA):**
Auto ARIMA provided the statistical benchmark, capturing trends and seasonality (52 weeks or 12 months). Models were selected using AIC and residual checks.

Machine Learning & Deep Learning:

- **XGBoost** used lagged features, rolling statistics, and seasonal inputs. Hyperparameters were tuned with expanding-window validation.
- **LSTM** leveraged sequential dependencies, trained on historical sales windows. Parameters such as units, layers, dropout, and learning rate were optimized via KerasTuner, with early stopping to prevent overfitting.

Hybrid Models:

- **Sequential Hybrid:** SARIMA modelled linear seasonality, while residuals were passed to LSTM to capture nonlinear patterns.
- **Parallel Hybrid:** SARIMA and LSTM were trained separately, and forecasts were combined through weighted averaging.

Monthly Prediction (8-Month Horizon):

- To assess higher-level patterns, both **XGBoost** and **Auto ARIMA** were retrained on monthly aggregated data. This reduced noise compared to weekly sales and highlighted seasonality. Accuracy was measured with MAE and MAPE, with ARIMA showing stable long-term structure and XGBoost adapting better to nonlinear fluctuations.

EXPLANATION OF VISUALISATIONS

1. Time Series Plots (Pre- and Post-2012):

Historical sales trends were visualised across the full timeline, with a clear split between pre-2012 and post-2012 data. This provided context on long-term patterns and highlighted how sales dynamics evolved after 2012.

2. STL Decomposition:

- Decompose the time series into trend, seasonal, and residual components.
- Helps to identify underlying patterns, seasonality, and irregular fluctuations.

3. ACF and PCF plots:

- plots show correlation of the series with its lagged values.
- Useful for selecting appropriate parameters for classical models.

4. Classical Approach – Auto ARIMA:

Forecast plots from Auto ARIMA illustrated the baseline performance of a statistical method, showing how well it captured trend and seasonality within weekly sales data.

5. Machine Learning Approach – XGBoost:

Forecast plots were created for both **weekly** and **monthly** horizons. Weekly plots showed responsiveness to short-term fluctuations, while monthly aggregation smoothed noise, making long-term trend predictions more interpretable.

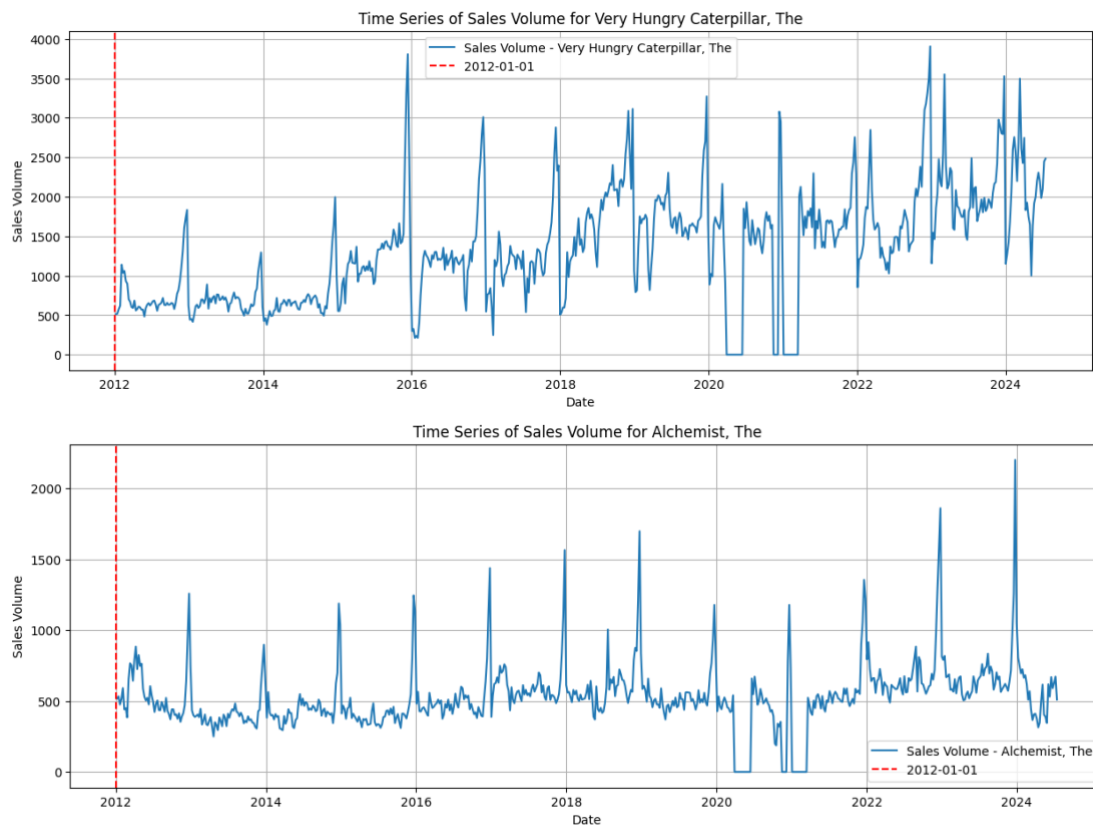
6. Deep Learning Approach – LSTM:

Multiple sets of plots were generated:

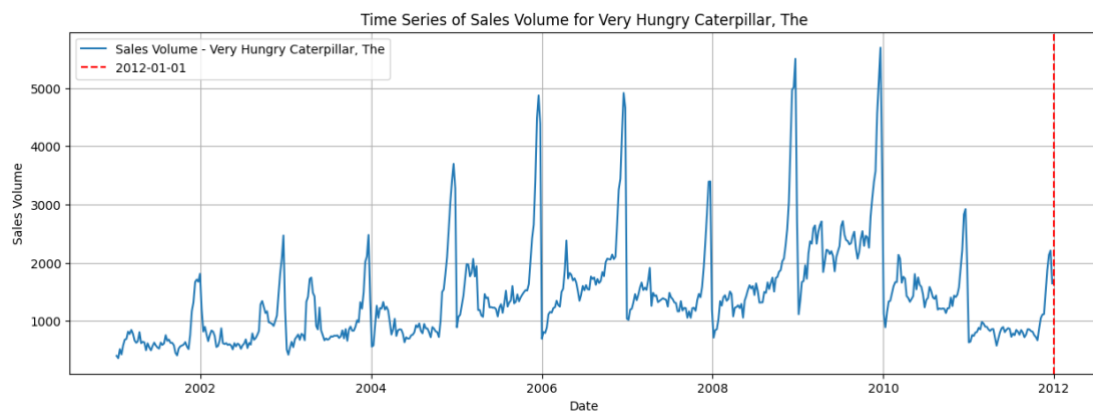
- **Simple LSTM** for baseline sequence learning,
- **Tuned LSTM** after hyperparameter optimisation,
- **Hybrid Models (SARIMA + LSTM):**
 - **Sequential Hybrid** plots where SARIMA outputs were passed into LSTM,
 - **Parallel Hybrid** plots where SARIMA and LSTM forecasts were combined via weighted averaging.

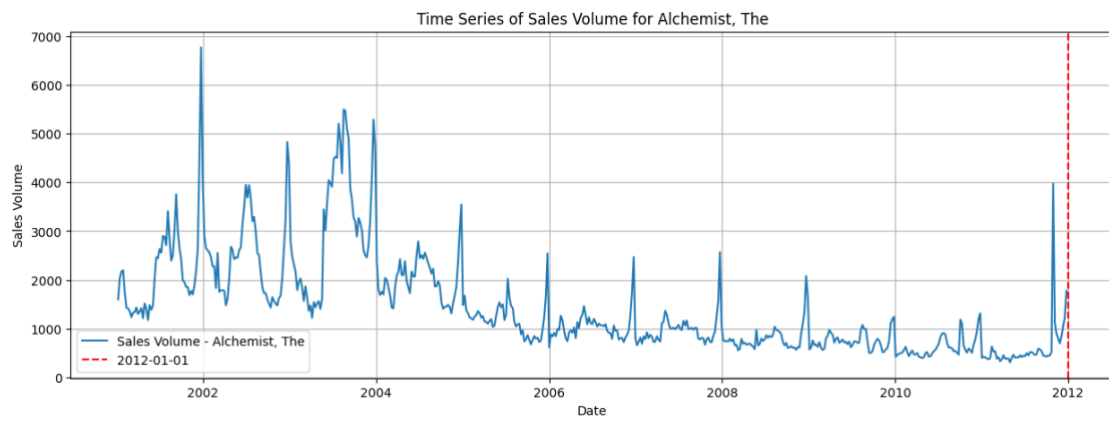
DATA VISUALISATION

1. Time Series Post-2012



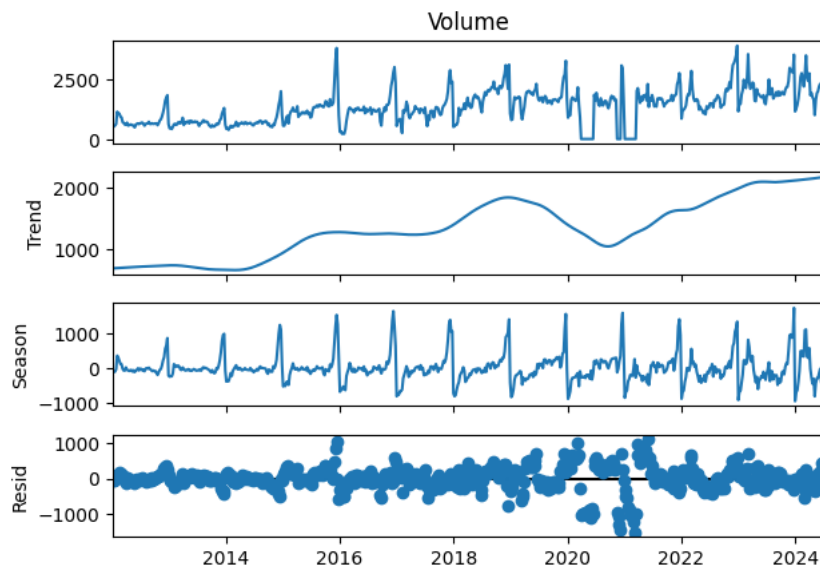
2. Time Series Pre-2012



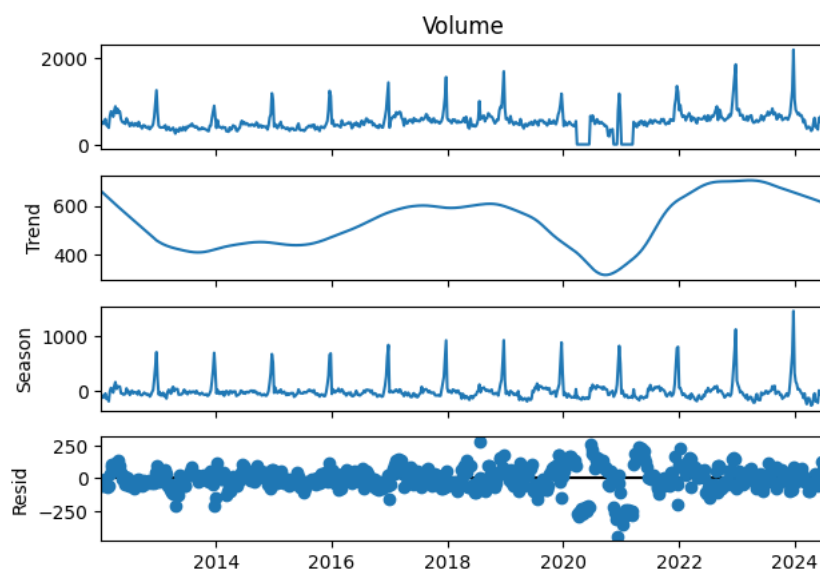


3. STL Decomposition

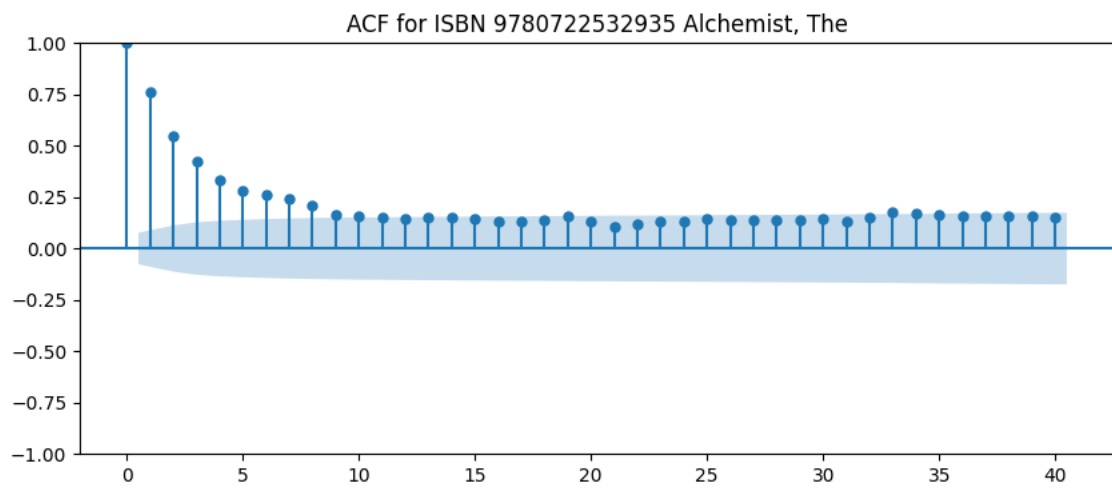
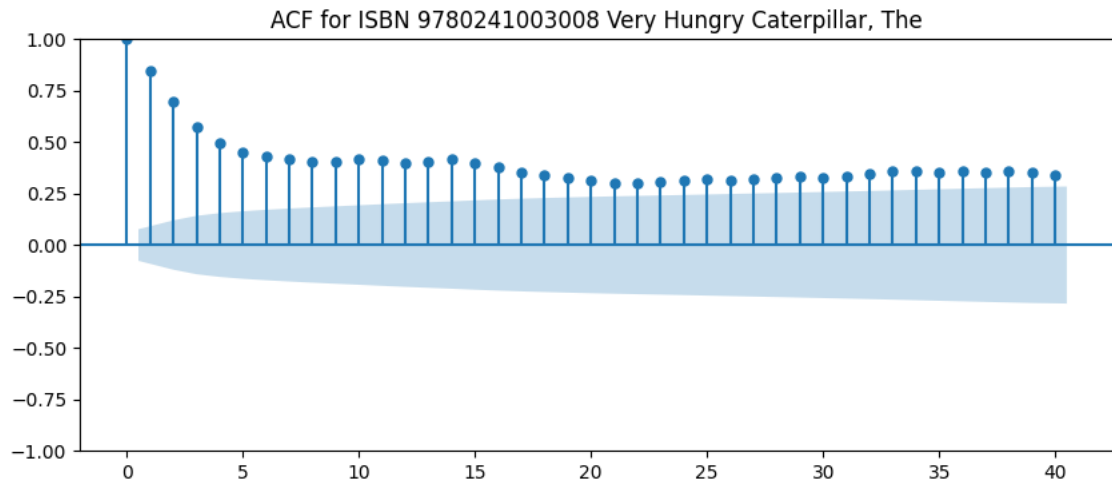
STL Decomposition - ISBN: 9780241003008 Very Hungry Caterpillar, The



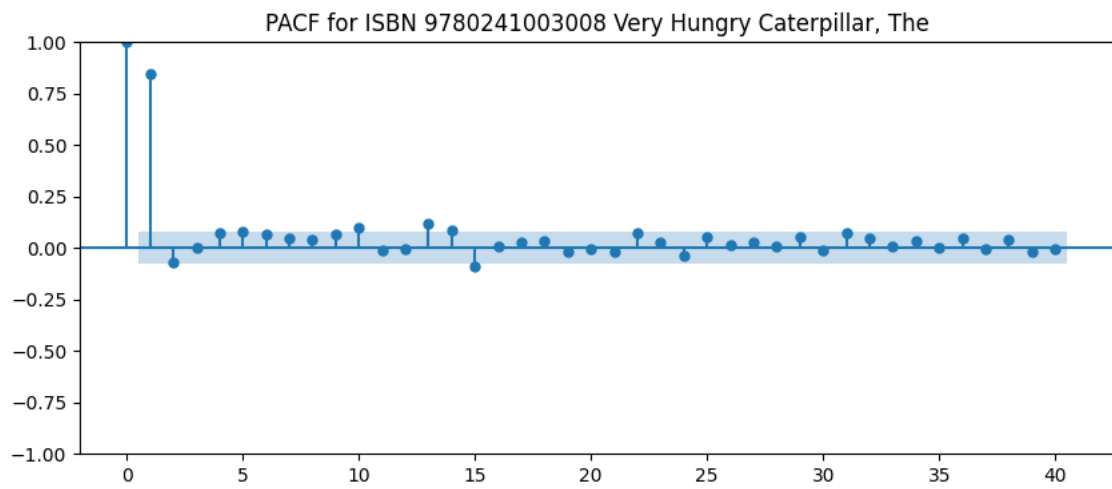
STL Decomposition - ISBN: 9780722532935 Alchemist, The

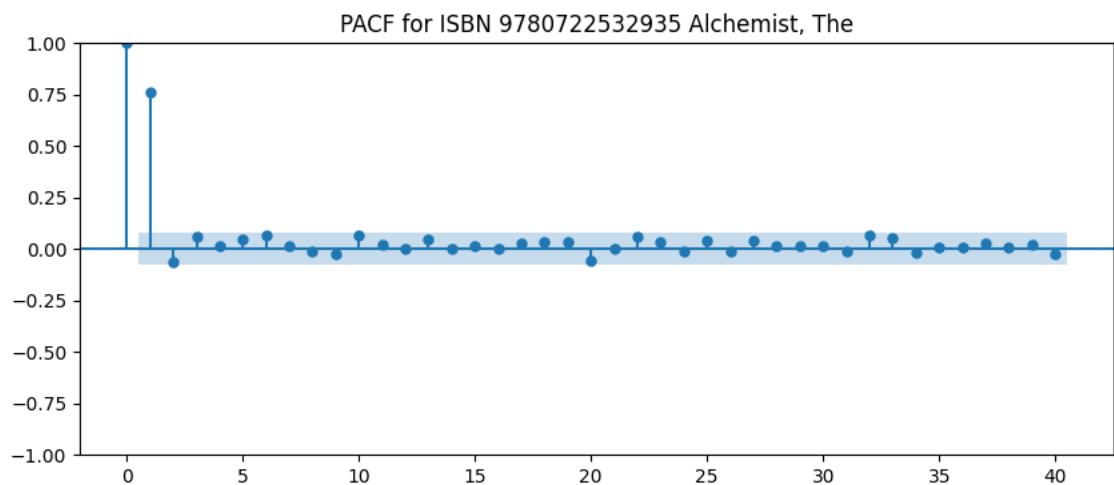


4. ACF Plots

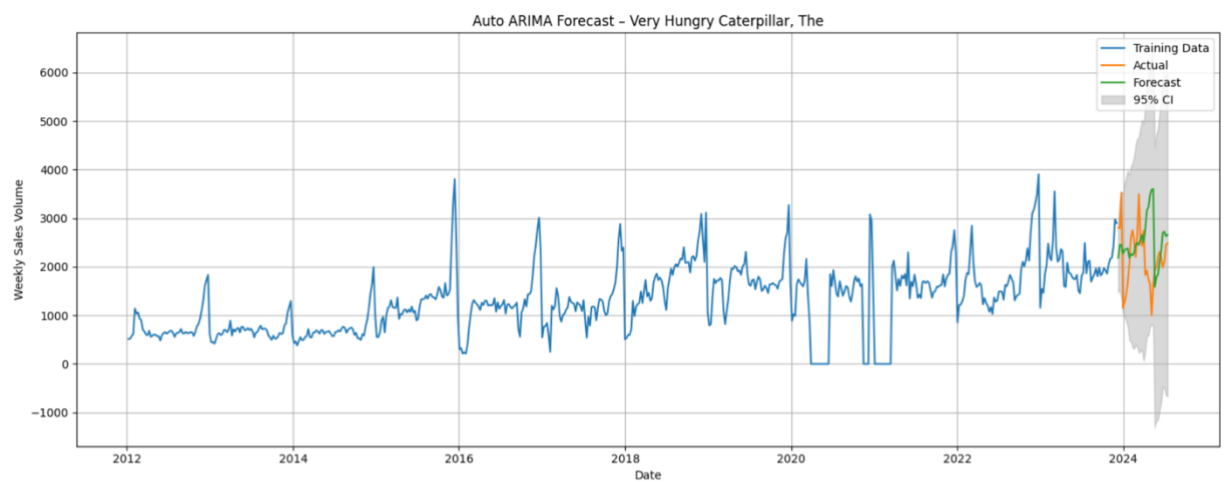


5. PACF Plots





6. Auto ARIMA(SARIMA)



Best model for Very Hungry Caterpillar, The:

SARIMAX Results

```

=====
Dep. Variable:          y          No. Observations:          654
Model:                SARIMAX(0, 1, 1)x(0, 1, 1, 52)      Log Likelihood          -4406.469
Date:                  Sat, 16 Aug 2025                    AIC              8818.937
Time:                  16:16:26                            BIC              8832.133
Sample:                01-08-2012                        HQIC              8824.074
                    - 07-14-2024
Covariance Type:      opg
=====

```

	coef	std err	z	P> z	[0.025	0.975]
ma.L1	-0.1823	0.031	-5.905	0.000	-0.243	-0.122
ma.S.L52	-0.5924	0.021	-28.853	0.000	-0.633	-0.552
sigma2	1.326e+05	2579.847	51.407	0.000	1.28e+05	1.38e+05

```

=====
Ljung-Box (L1) (Q):          0.44      Jarque-Bera (JB):          7373.73
Prob(Q):                    0.51      Prob(JB):              0.00
Heteroskedasticity (H):      6.29      Skew:                  1.59
Prob(H) (two-sided):         0.00      Kurtosis:              19.86
=====

```

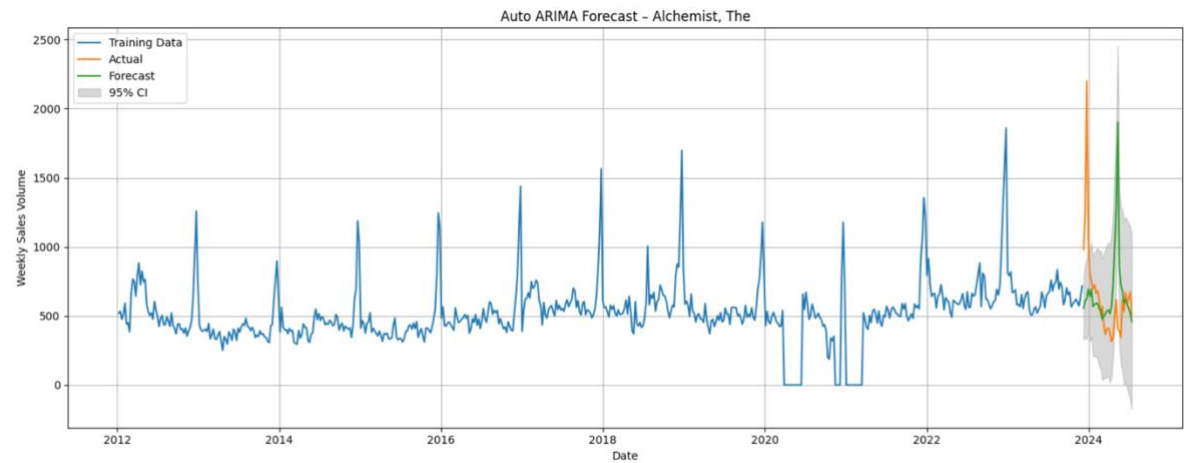
Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).

Mean Squared Error: 843337.7623

Mean Absolute Error: 689.4329

Mean Absolute Percentage Error: 40.57%



Best model for Alchemist, The:

SARIMAX Results

```

Dep. Variable:          y          No. Observations:      654
Model:                SARIMAX(0, 1, 2)x(1, 1, [1], 52)    Log Likelihood      -3714.981
Date:                 Sat, 16 Aug 2025                    AIC                7439.962
Time:                 16:23:30                             BIC                7461.955
Sample:               01-08-2012                         HQIC               7448.522
                  - 07-14-2024
Covariance Type:      opg

```

	coef	std err	z	P> z	[0.025	0.975]
ma.L1	-0.3648	0.028	-12.922	0.000	-0.420	-0.309
ma.L2	-0.1740	0.035	-4.975	0.000	-0.242	-0.105
ar.S.L52	-0.1958	0.075	-2.608	0.009	-0.343	-0.049
ma.S.L52	-0.2379	0.069	-3.431	0.001	-0.374	-0.102
sigma2	1.348e+04	380.339	35.446	0.000	1.27e+04	1.42e+04

```

Ljung-Box (L1) (Q):      0.01  Jarque-Bera (JB):      2163.24
Prob(Q):                 0.91  Prob(JB):              0.00
Heteroskedasticity (H):   3.72  Skew:              1.25
Prob(H) (two-sided):      0.00  Kurtosis:         11.95

```

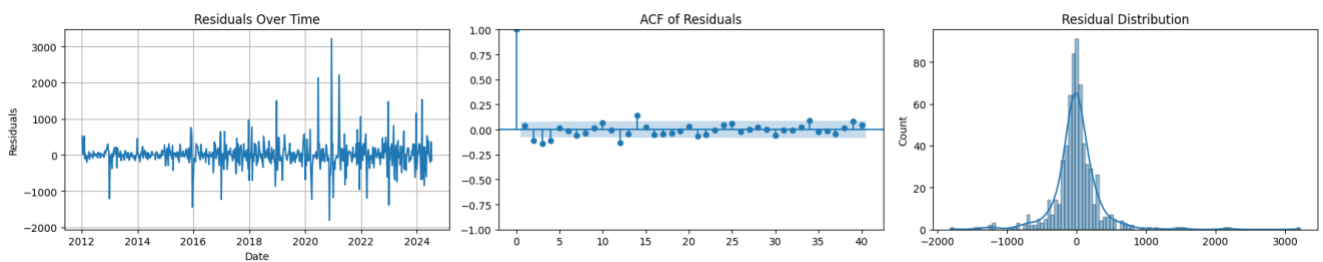
Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).

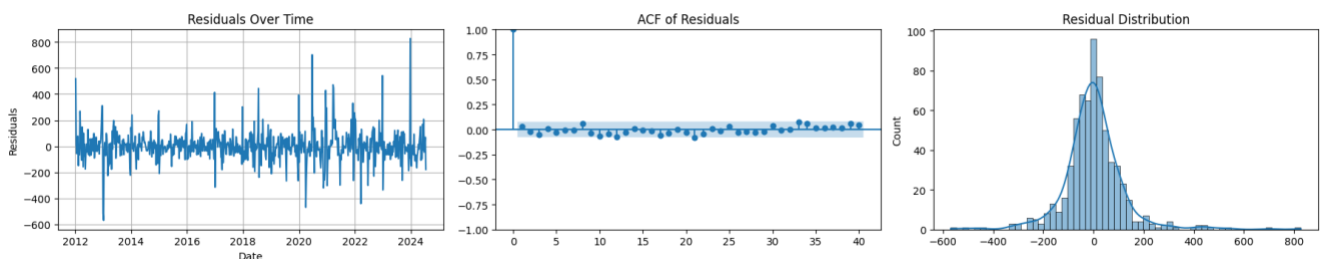
Mean Squared Error: 218650.2358
Mean Absolute Error: 285.1041
Mean Absolute Percentage Error: 48.92%

7. Auto ARIMA Residuals for best models

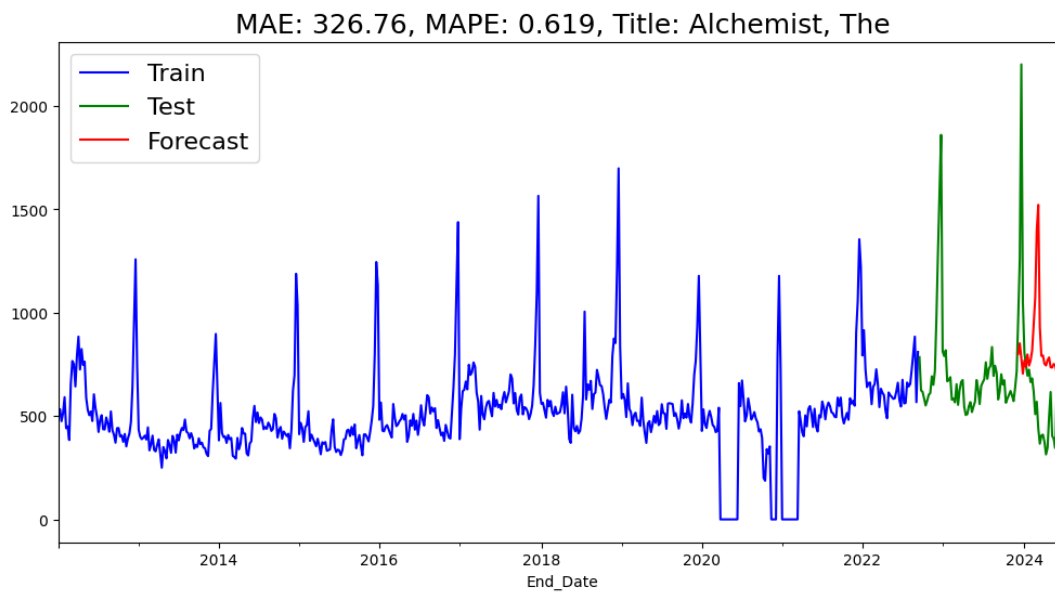
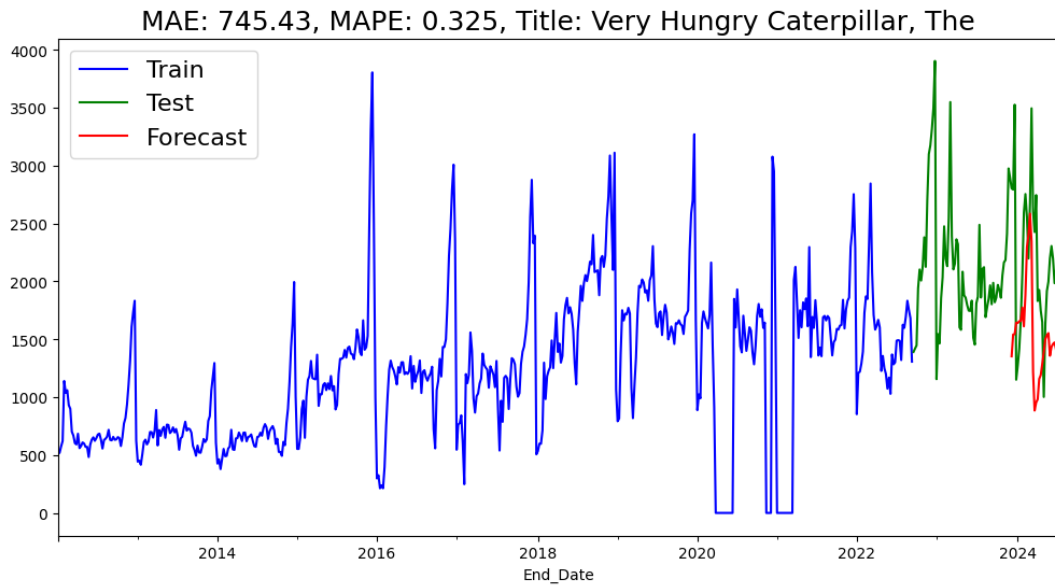
Residual Diagnostics - ISBN 9780241003008 - Very Hungry Caterpillar, The



Residual Diagnostics - ISBN 9780722532935 - Alchemist, The

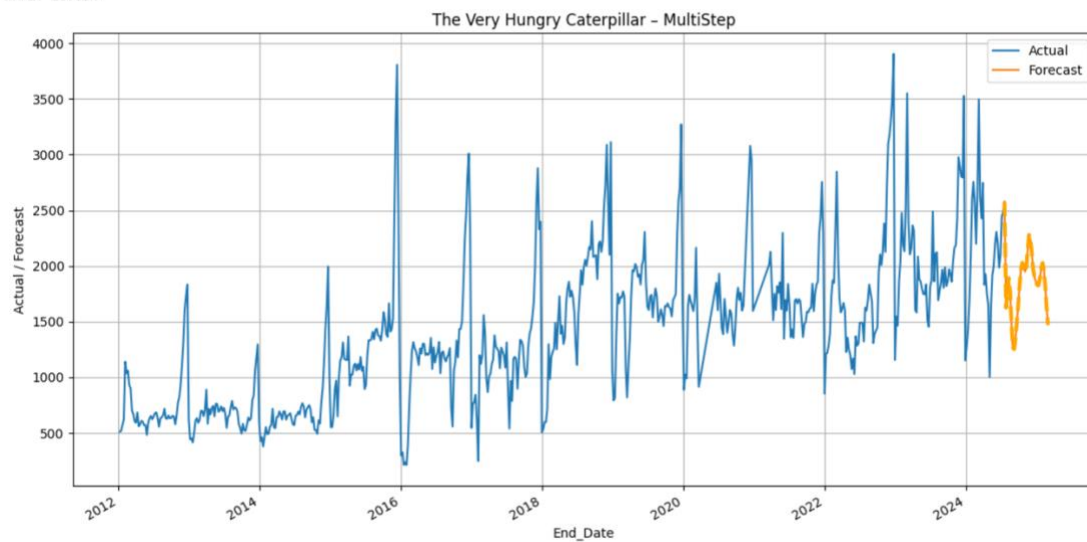


8. XG Boost Pipeline

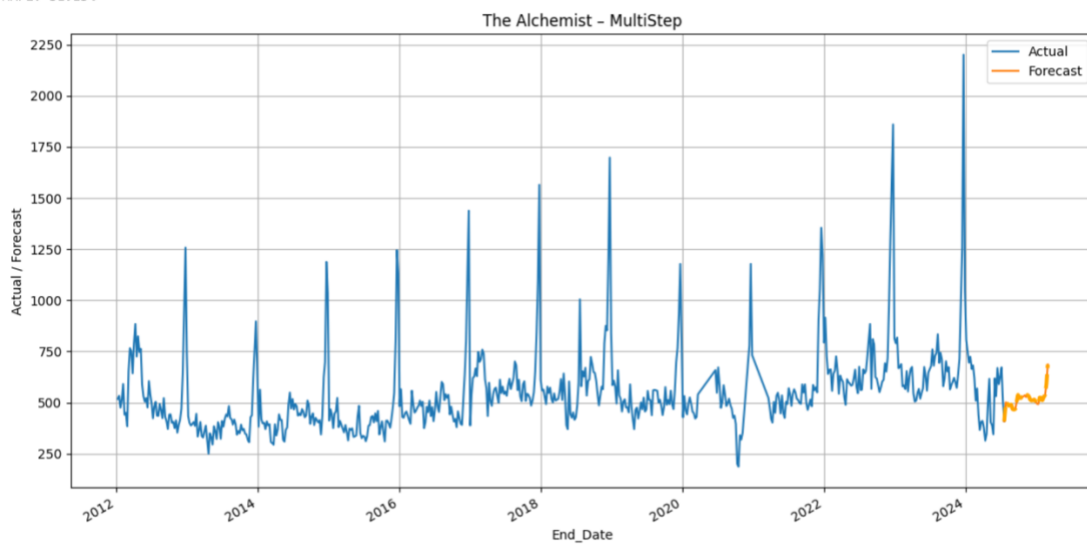


9. LSTM Multistep

MAE: 629.12
MAPE: 28.83%

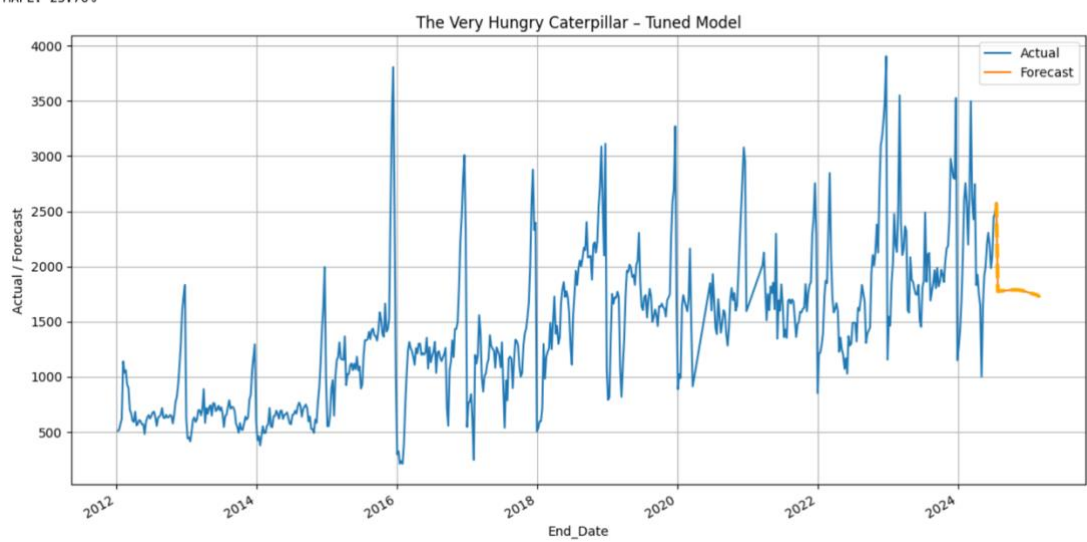


MAE: 221.73
MAPE: 31.13%

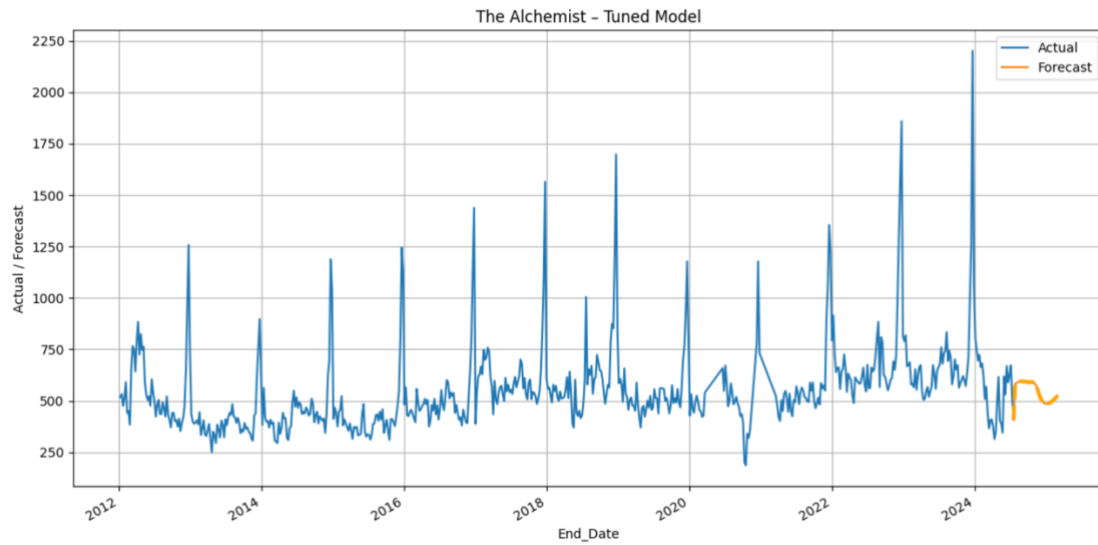


10. LSTM Tuned

MAE: 588.64
MAPE: 25.76%



MAE: 195.64
MAPE: 28.33%



11. Hybrid Model (Sequential)

SARIMAX Results

```
=====
Dep. Variable:          Volume      No. Observations:      604
Model:                 SARIMAX(1, 1, 1)x(1, 0, 1, 12)  Log Likelihood      -4360.124
Date:                  Sun, 17 Aug 2025  AIC              8732.248
Time:                  19:32:16      BIC              8758.659
Sample:                0              HQIC             8742.527
Covariance Type:      opg
=====
```

	coef	std err	z	P> z	[0.025	0.975]
intercept	-1.4571	4.402	-0.331	0.741	-10.084	7.170
ar.L1	0.7222	0.023	31.544	0.000	0.677	0.767
ma.L1	-1.0000	0.458	-2.181	0.029	-1.898	-0.102
ar.S.L12	-0.9999	6.132	-0.163	0.870	-13.018	11.018
ma.S.L12	0.9998	6.588	0.152	0.879	-11.913	13.913
sigma2	1.108e+05	9.91e-06	1.12e+10	0.000	1.11e+05	1.11e+05

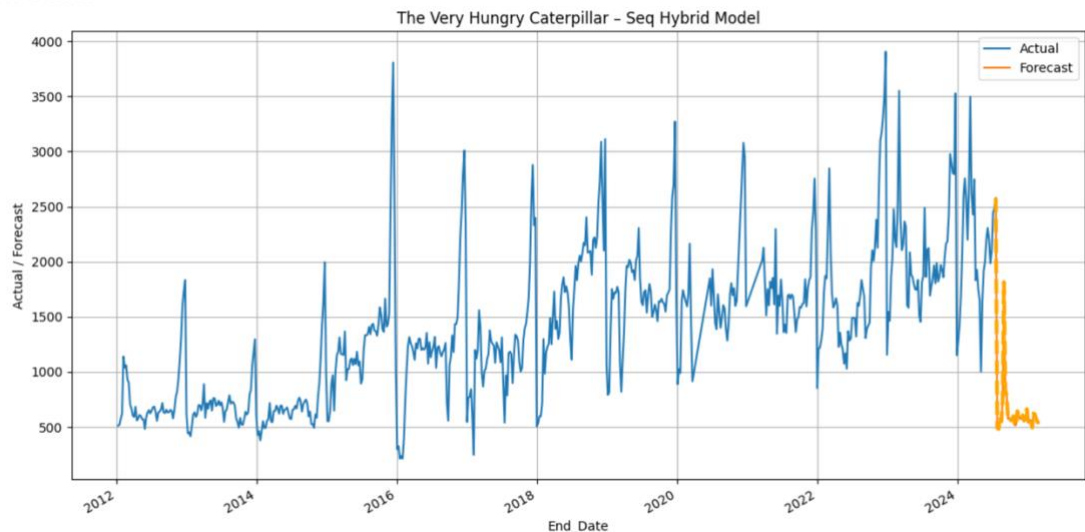
```
=====
Ljung-Box (L1) (Q):      2.78      Jarque-Bera (JB):      5912.47
Prob(Q):                 0.10      Prob(JB):              0.00
Heteroskedasticity (H):  0.36      Skew:                  2.75
Prob(H) (two-sided):     0.00      Kurtosis:              17.32
=====
```

Warnings:

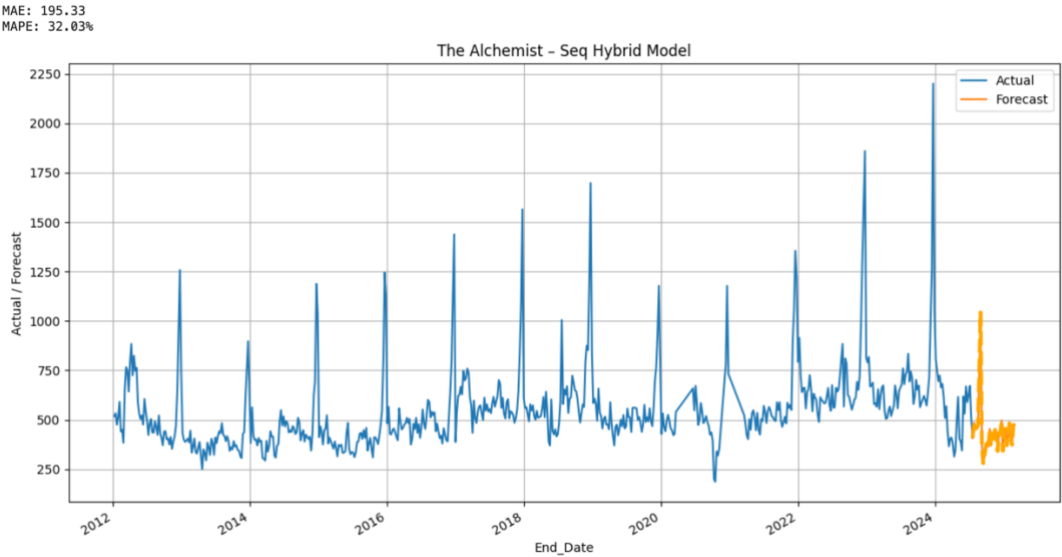
[1] Covariance matrix calculated using the outer product of gradients (complex-step).

[2] Covariance matrix is singular or near-singular, with condition number 2.35e+28. Standard errors may be unstable.

MAE: 158.04
MAPE: 20.51%

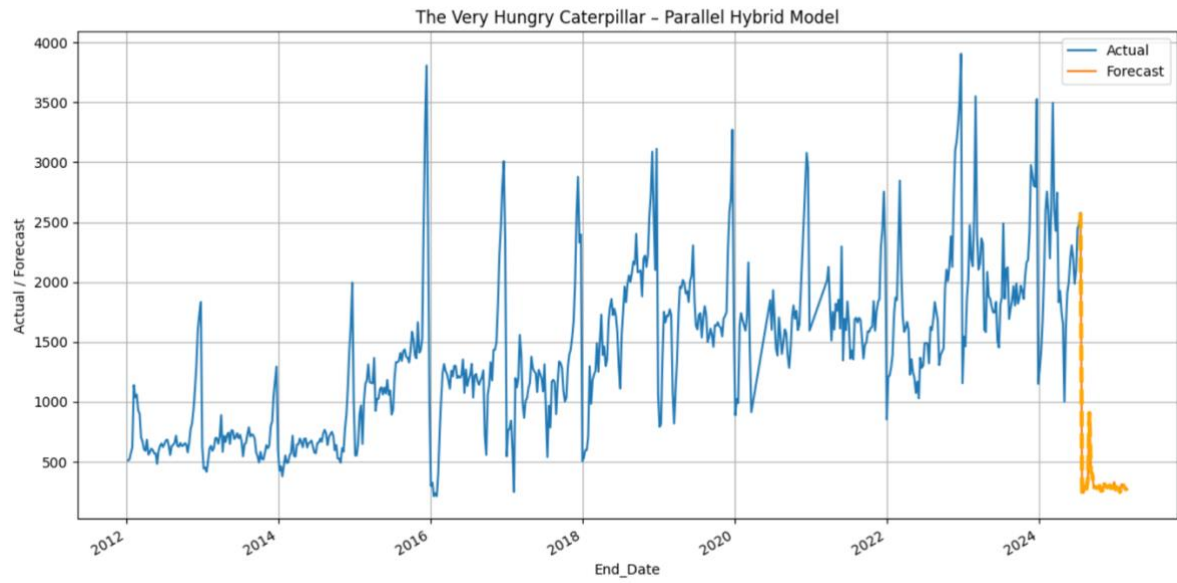


SARIMAX Results						
Dep. Variable:	Volume	No. Observations:	604			
Model:	SARIMAX(1, 1, 1)x(1, 0, 1, 12)	Log Likelihood	-3838.199			
Date:	Sun, 17 Aug 2025	AIC	7688.398			
Time:	19:32:20	BIC	7714.809			
Sample:	0	HQIC	7698.677			
	- 604					
Covariance Type:	opg					
	coef	std err	z	P> z	[0.025	0.975]
intercept	-0.0056	0.039	-0.144	0.886	-0.083	0.071
ar.L1	0.6823	0.019	35.770	0.000	0.645	0.720
ma.L1	-1.0000	1.324	-0.755	0.450	-3.595	1.595
ar.S.L12	0.9584	0.291	3.299	0.001	0.389	1.528
ma.S.L12	-0.9673	0.270	-3.586	0.000	-1.496	-0.439
sigma2	1.966e+04	2.59e+04	0.759	0.448	-3.11e+04	7.04e+04
Ljung-Box (L1) (Q):	2.04	Jarque-Bera (JB):	24604.78			
Prob(Q):	0.15	Prob(JB):	0.00			
Heteroskedasticity (H):	0.45	Skew:	4.36			
Prob(H) (two-sided):	0.00	Kurtosis:	33.05			
Warnings:						
[1] Covariance matrix calculated using the outer product of gradients (complex-step).						

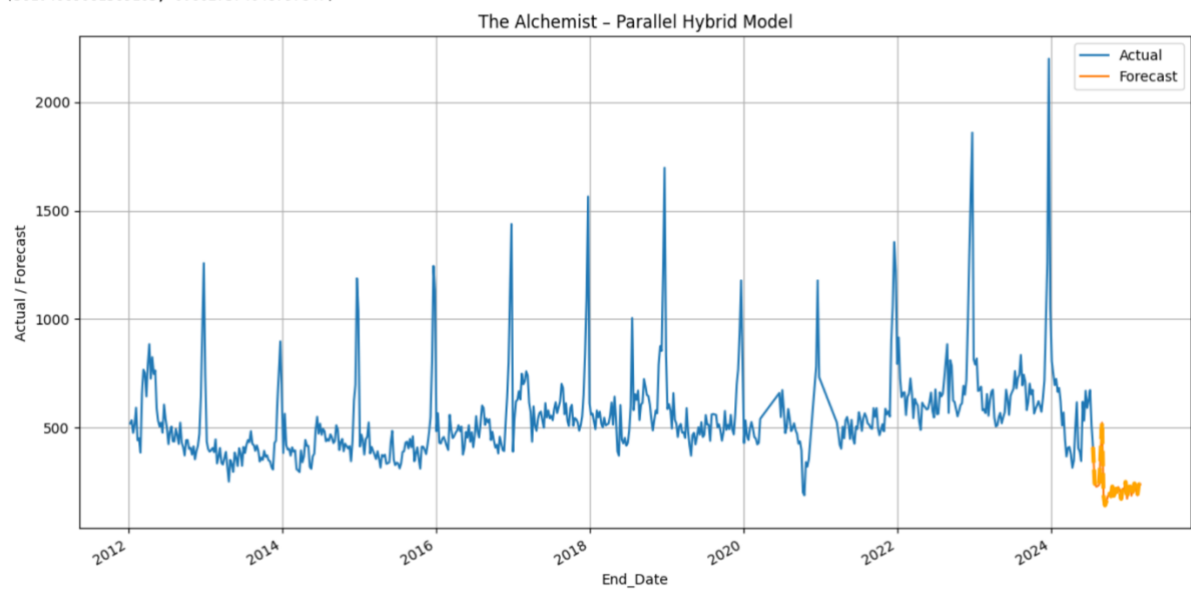


12. Hybrid Model (Parallel) Weightage (LSTM 50% - SARIMA 50%)

MAE: 372.43
 MAPE: 53.58%
 (372.4297813630517, 0.5357883435318471)

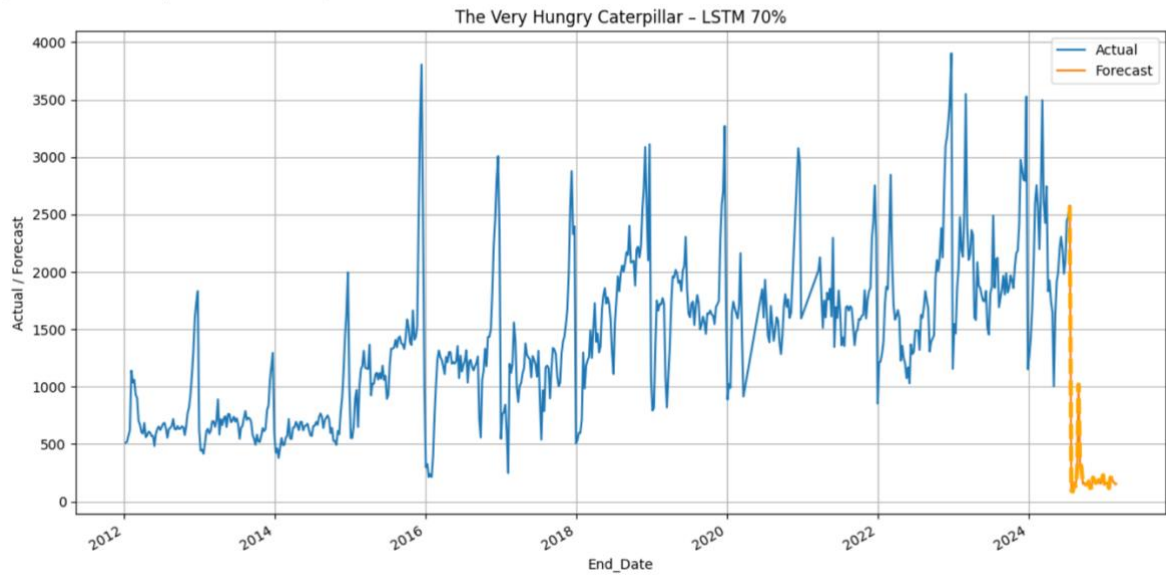


MAE: 361.41
 MAPE: 60.27%
 (361.4069061369105, 0.6027374048797847)

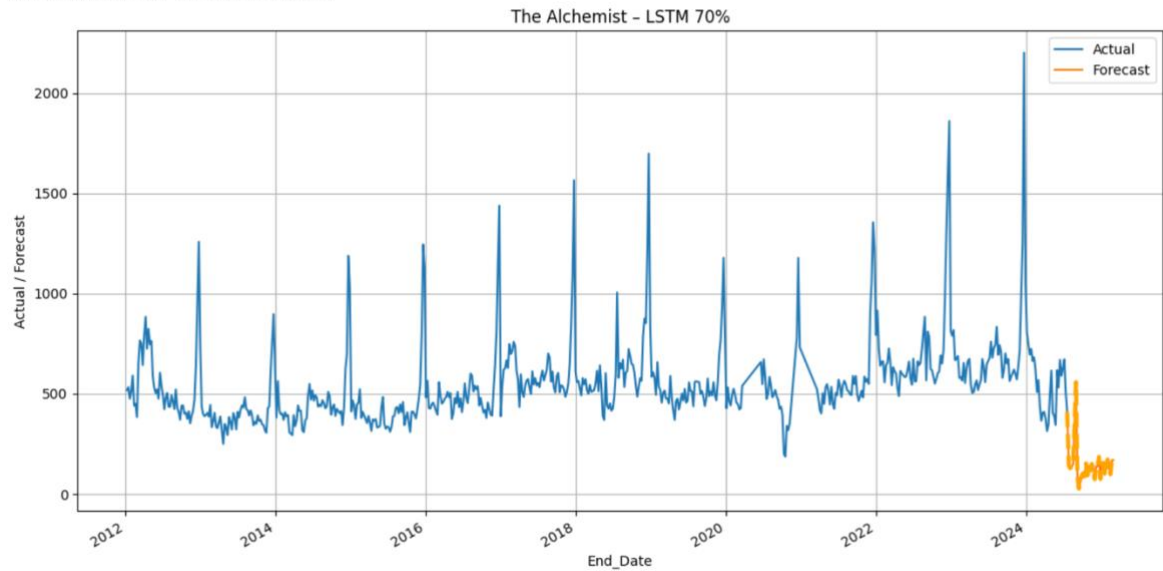


13. Hybrid Model (Parallel) Weightage (LSTM 70% - SARIMA 30%)

MAE: 494.05
 MAPE: 72.56%
 (494.04557093796745, 0.7255872181633075)

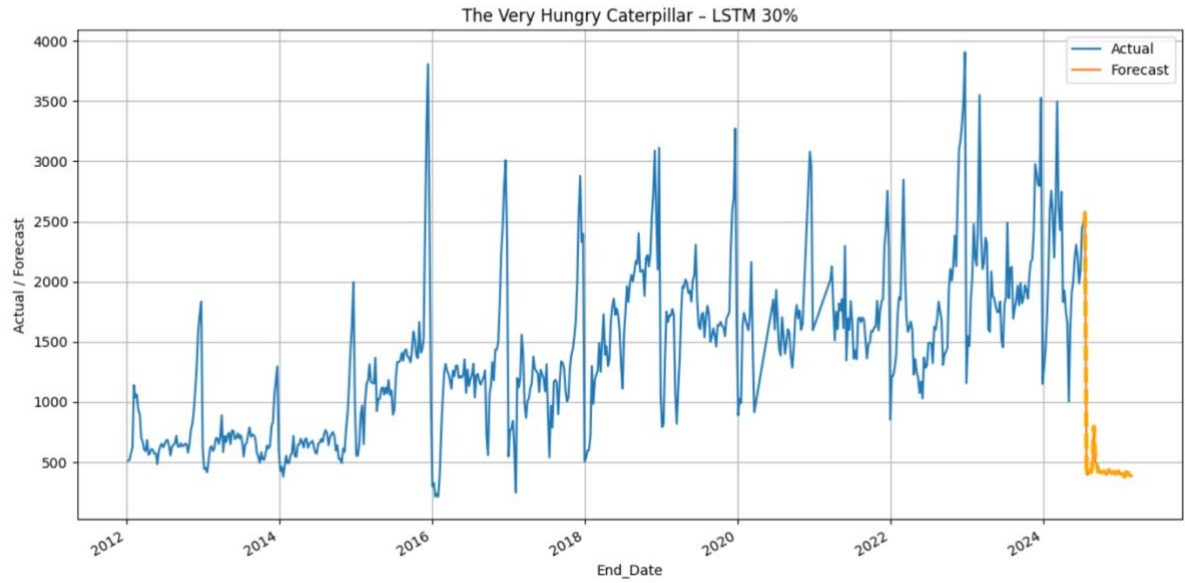


MAE: 442.89
 MAPE: 75.01%
 (442.88614726575406, 0.7501378697479355)

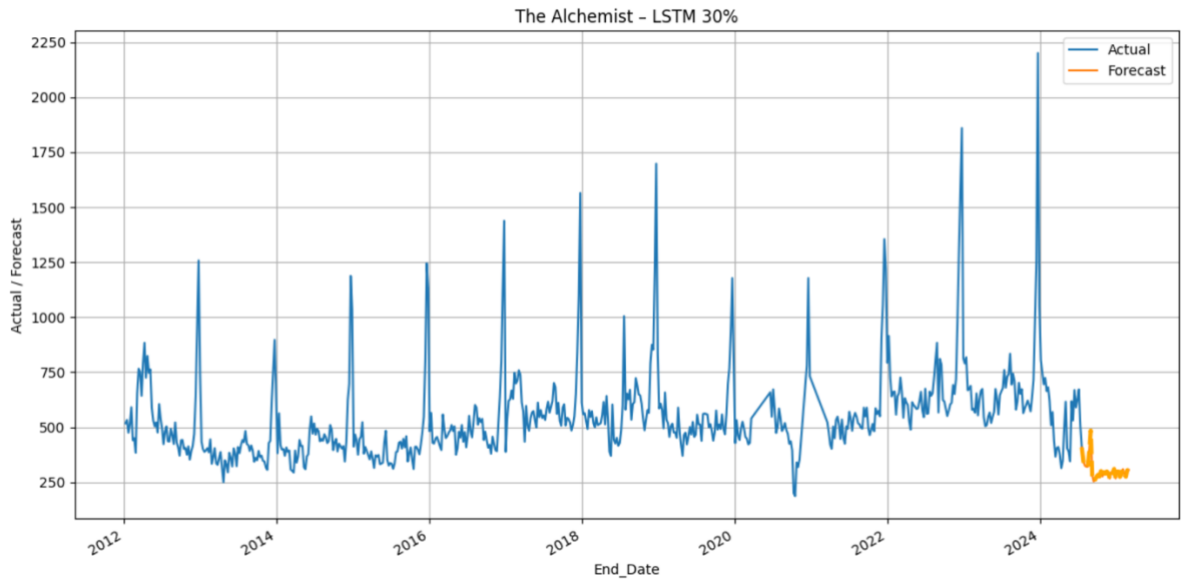


14. Hybrid Model (Parallel) Weightage (LSTM 30% - SARIMA 70%)

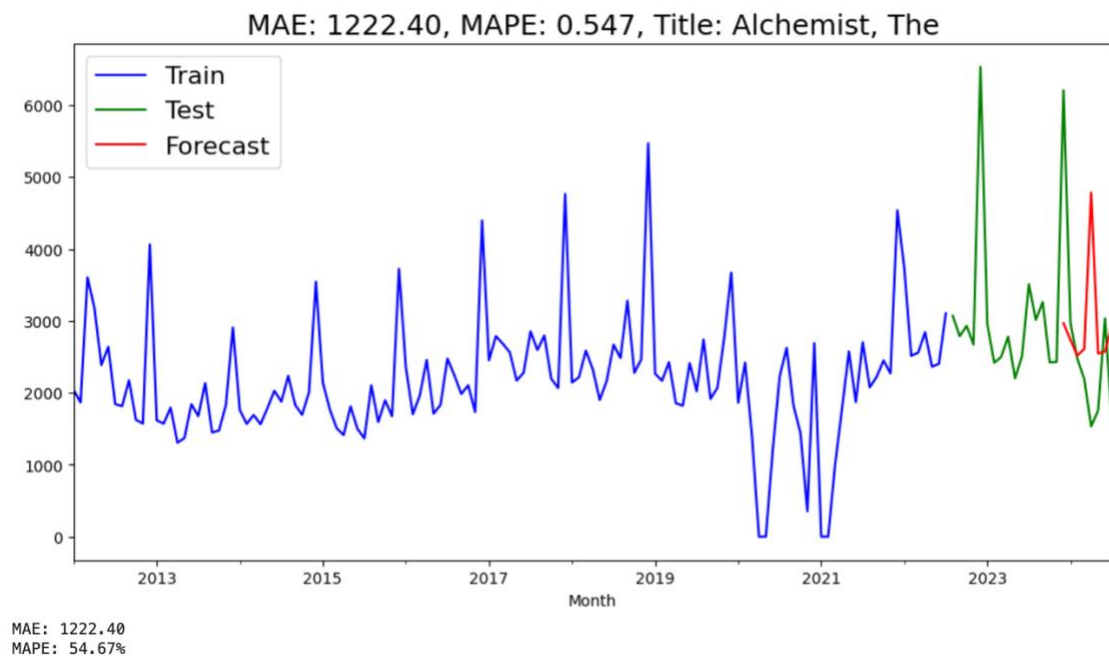
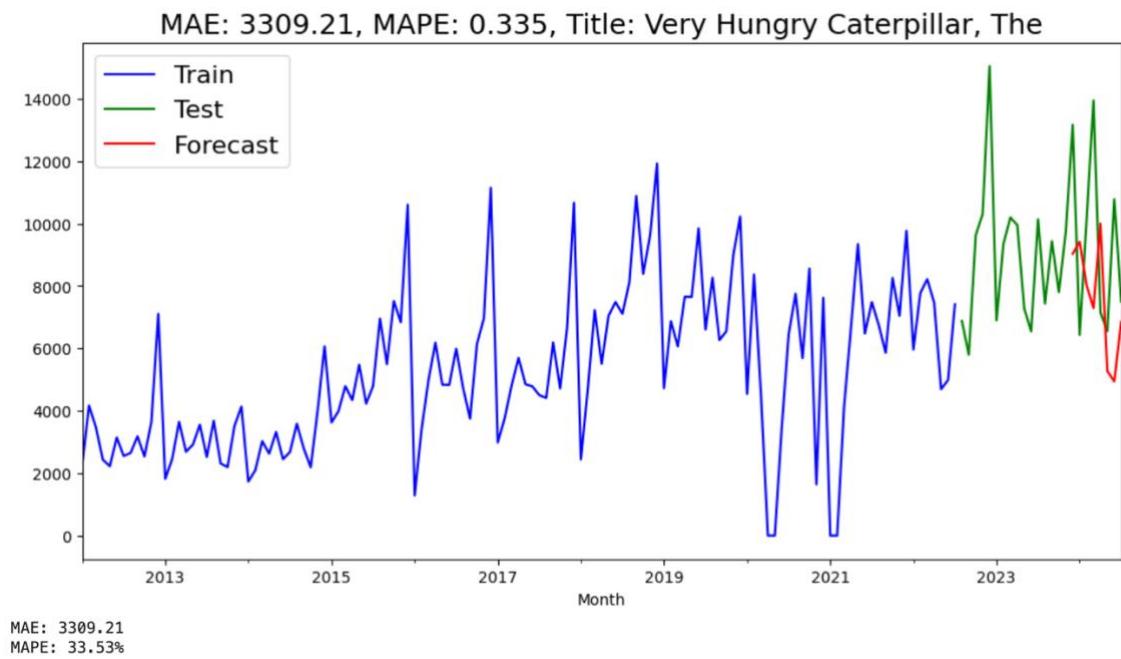
MAE: 250.81
MAPE: 34.60%
(250.81399178813592, 0.34598946890038673)



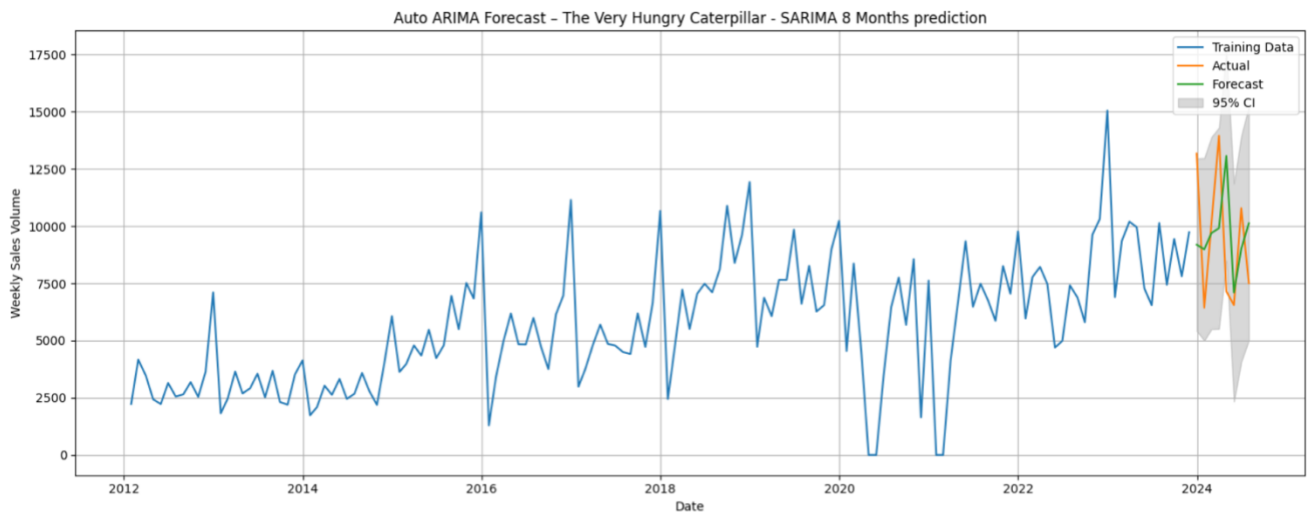
MAE: 279.94
MAPE: 45.54%
(279.9356261781366, 0.45535114674006943)



15. XGBoost Monthly Prediction



16. SARIMA (Auto-ARIMA)



Best model for The Very Hungry Caterpillar - SARIMA 8 Months prediction:
SARIMAX Results

```
=====
Dep. Variable:          y      No. Observations:      151
Model:                 SARIMAX(0, 1, 1)x(0, 1, 1, 12)  Log Likelihood    -1245.102
Date:                  Sun, 17 Aug 2025               AIC             2496.203
Time:                  19:33:13                       BIC             2504.985
Sample:                01-31-2012                     HQIC            2499.772
                  - 07-31-2024
Covariance Type:      opg
=====
```

	coef	std err	z	P> z	[0.025	0.975]
ma.L1	-0.6530	0.050	-13.040	0.000	-0.751	-0.555
ma.S.L12	-0.8641	0.092	-9.385	0.000	-1.045	-0.684
sigma2	3.675e+06	4.08e+05	9.004	0.000	2.88e+06	4.48e+06

```
=====
Ljung-Box (L1) (Q):                2.67  Jarque-Bera (JB):                14.26
Prob(Q):                           0.10  Prob(JB):                      0.00
Heteroskedasticity (H):             3.18  Skew:                          -0.37
Prob(H) (two-sided):                0.00  Kurtosis:                      4.39
=====
```

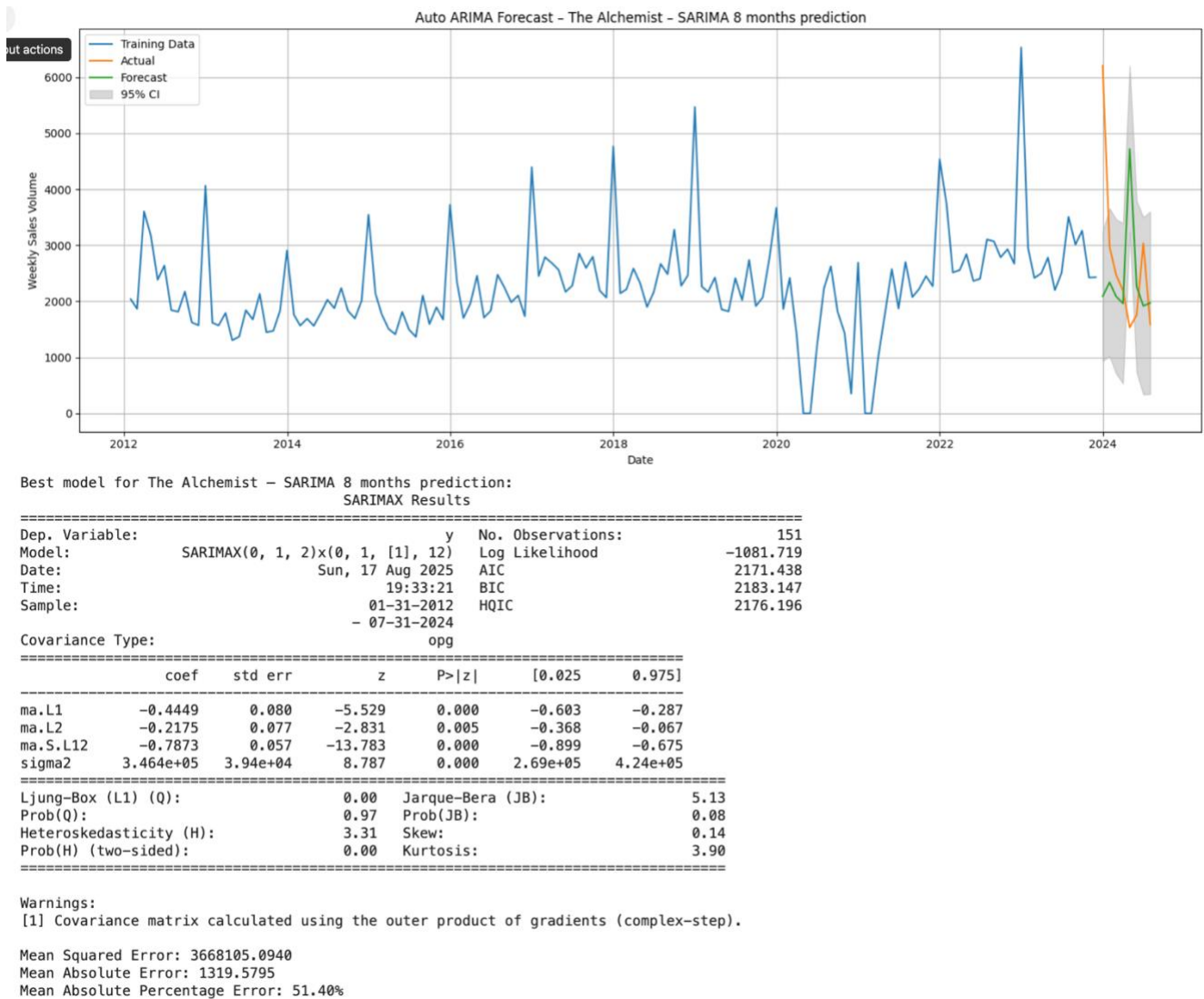
Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).

Mean Squared Error: 10519996.4380

Mean Absolute Error: 2730.1850

Mean Absolute Percentage Error: 30.67%



INSIGHTS

1. Aggregation (weekly vs monthly).

- Weekly models tend to excel when there are short-lived shocks—promotions, school holidays, media mentions—that don't align neatly with month boundaries.
- Monthly models (Auto ARIMA) benefit from noise reduction due to aggregation. If a title's demand pattern is smoother (steady backlist sales, predictable seasonal peaks).

2. Computation & practicality

- a. Auto ARIMA is fast at monthly frequency ($m=12$), while weekly Auto ARIMA with $m=52$ can be heavier.
- b. Multi step LSTM models used 600 epochs for better learning, which takes considerable amount of time while the Tuned models with Hyper Parameters used 200 epochs.

3. Accuracy

The most accurate models have been listed below in decreasing order of accuracy:

- Parallel Hybrid (SARIMA + LSTM) (Weekly)
- Sequential Hybrid (Weekly)
- Tuned LSTM (Weekly)
- XGBoost and Auto-ARIMA(SARIMA)(monthly)

The LSTM can overreact to local patterns; SARIMA can underreact during sudden changes. Weighted averaging yielded forecasts that were **smoother** than LSTM alone but **more adaptive** than SARIMA alone, improving error metrics.

CONCLUSION

The **Parallel Hybrid SARIMA + LSTM** model balances the strengths of statistical and deep learning approaches, providing more accurate and stable forecasts for book sales, especially over the final 32 weeks or 8-month monthly horizons.

INFERENCES

The analysis shows that book sales exhibit strong seasonal and trend patterns. Classical models like SARIMA capture seasonality effectively, while XGBoost and LSTM detect nonlinear trends and sudden changes. Hybrid models, combining SARIMA with LSTM (sequential or parallel), consistently outperform single approaches. Weekly forecasts are more volatile, whereas monthly aggregation smooths noise and improves stability. Short-term forecasts can rely on XGBoost or tuned LSTM, but long-term strategic decisions benefit from hybrid models.